

16/10/12

Code: MCS101

LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING (AUTONOMOUS)

L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Supplementary Examinations, October 2012

DATA STRUCTURES & ALGORITHM ANALYSIS

(Computer Science and Engineering)

Time: 3 hours

Max. Marks: 60

Answer all the questions

All Questions carry equal marks

- 1(a) Explain Different types of asymptotic notation's [6M]
(b) Explain stack operations [6M]

(OR)

- (c) Explain representation and implementation of circular linked lists [12M]
2(a) Explain merge sort algorithm with an example. Mention its complexity.
(b) Explain Heapsort algorithm [6+6M]

(OR)

- (c) Write short notes on hash functions and Rehashing [12M]
3(a) Define Binary tree. Draw binary tree for the following expression
(a+b+c)+(cd*e+f) * g [6M]
(b) Explain single rotation on AVL trees. [6M]

(OR)

- (c) Explain DFS algorithm with an example. [6M]
(d) Write short notes on biconnected components. [6M]
4(a) Explain Dijkstra's algorithm with an example. [6M]
(b) Explain Kruskal's algorithm with an example. [6M]

(OR)

- (c) Write control abstraction for divide and conquer technique [6M]
(d) Write algorithm for sequencing jobs with deadline. [6M]
5(a) Write an algorithm for finding shortest path using all pairs. [6M]
(b) Explain 0/1 knapsack problem. [6M]

(OR)

- (c) Explain travelling sales persons problem. [12M]

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Code: MCS103

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M.Tech. I Semester Supplementary Examinations, October 2012

COMPUTER ORGANIZATION

(Computer Science and Engineering)

Time: 3 hours

Max. Marks: 60

Answer all the questions

All Questions carry equal marks

- 1(a) Design the bus system for four registers with three state buffer gate
(b) Explain different types of shifts. [6+6M]

(OR)

- (c) List out the various Arithmetic micro operations. [4M]
(d) Explain how add micro operation is implemented. [8M]

- 2(a) Explain the functions of various registers in a basic computer [6M]
(b) Describe the control unit of a basic computer system. [6M]

(OR)

- (c) Describe address sequencing. [6M]
(d) Describe microprogram sequencer for a control memory. [6M]

- 3(a) Describe instruction formats. [6M]
(b) Describe the major characteristics of a RISC processor. [6M]

(OR)

- (c) Describe signed magnitude division operation with a flow chart. [6M]
(d) Describe addition and subtraction operation operations of floating point numbers with a neat flow chart. [6M]

- 4(a) Explain main memory in detail [6M]
(b) Describe associative memory [6M]

(OR)

- (c) Explain address mapping using pages [6M]
(d) Explain memory management hardware [6M]

- 5(a) Describe I/O interface unit [6M]
(b) Explain Handshaking [6M]

(OR)

- (c) Describe programmed I/O and interrupt initiate I/O [6M]
(d) Explain serial communication [6M]

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M.Tech. I Semester Supplementary Examinations, October 2012

AUTOMATA AND COMPILER DESIGN

(Computer Science and Engineering)

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal marks

- 1(a) Prove that for every NDFA, there exists a DFA which simulates the behavior of NDFA [12M]

(OR)

- (b) Write a short note on predicative parsing [6M]
(c) Write a short note on lexical analyzer [6M]

- 2(a) Show that the grammar

$S \rightarrow a/abSb/aAb$

$A \rightarrow bS/aAAb$ is ambiguous [6M]

- (b) Write a short note on backtracking [6M]

(OR)

- (c) For the grammar $S \rightarrow aAS/a$, $A \rightarrow SbA/SS/ba$. Construct derivation trees for the string aabbaa. [6M]

- (d) Write the rules for computing FIRST and FOLLOW [6M]

- 3(a) Write about LR parsing and LALR parsing [6M]

- (b) Write error recovery in LR parsing [6M]

(OR)

- (c) Write a short note on LEX [6M]

- (d) Write Error Recovery in YACC [6M]

- 4(a) Explain syntax-Directed translation of case statements. [6M]

- (b) Write a short note on storage organization [6M]

(OR)

- (c) Briefly explain rules for types checking [6M]

- (d) Write about quadruples and indirect triples. [6M]

- 5(a) Write a short note on Representation of Array references [6M]

- (b) Write a short note on register Allocation and Assignment [6M]

(OR)

- (c) Write the algorithm for constructing a DAG. What is the useful information that can be obtained by constructing a DAG. [12M]

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M.Tech. I Semester Supplementary Examinations, October 2012

VLSI TECHNOLOGY AND DESIGN

(Signals and System Processing)

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal marks

- 1(a) Discuss the origin of Latchup problems in CMOS circuits with necessary diagrams. Explain the remedial measures. [8M]
(b) Compare CMOS and BiCOMS technologies. [4M]
(OR)
(c) Explain CMOS inverter voltage transfer characteristic with a neat diagram. What is the criteria for voltage threshold for high level and low level in inverter characteristics? [8M]
(d) Explain the operation of pass transistors. [4M]
- 2(a) What is a stick diagram? What do you mean by Lambda based design rules. [6M]
(b) Distinguish between switch logic and gate logic. [6M]
(OR)
(c) Explain about resistive and inductive interconnect delays in gates? [6M]
(d) Explain the CMOS inverter layout plan along with its cross-sectional diagram. [6M]
- 3(a) Explain how fanout and path delay influences delay in a combinational networks. [6M]
(b) Design a layout diagram for a CMOS NOR gate? [6M]
(OR)
(c) Explain various list of simulation parameters in a switch logic networks. [6M]
(d) Explain the testing procedure for faults in combinational networks. [6M]
- 4(a) Give the routing techniques to equalize channel utilization. [6M]
(b) Explain about forming of array of memory cells. [6M]
(OR)
(c) Explain about clocked CMOS logic? [6M]
(d) Compare chip level testing and system level testing. [6M]
- 5(a) Write notes on Architecture testing of VLSI design. [6M]
(b) Explain about off chip connections. [6M]
(OR)
(c) What are the various floor planning methods? Explain them clearly? [6M]
(d) Explain system-on-chip concept using platform based design. [6M]

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L.B. Reddy Nagar :: Mylavaram – 521 230 :: Krishna Dist.

M.Tech. I Semester Supplementary Examinations, October 2012

MICROCONTROLLERS FOR EMBEDDED SYSTEM DESIGN

(Signals and System Processing)

Time: 3 hours

Max. Marks: 60

Answer all the questions

All Questions carry equal marks

- 1(a) Define Embedded system and explain about Embedded hardware units? [6M]
(b) Explain the skills and challenges in designing an Embedded system? [6M]

(OR)

- (c) Explain the classification of Embedded systems and write the applications of it? [6M]
(d) Write about (i) Embedded software (ii) Complex system Design. [6M]

- 2(a) Explain about Registers used in micro-controllers with diagrams? [6M]
(b) Explain about memory interfacing and memory controller? [6M]

(OR)

- (c) Draw the Interfacing 8051 Architecture and explain about Input/output ports? [8M]
(d) Write about the importance of counters and Timers in 8051 controller? [4M]

- 3(a) Draw the ARM processor architecture and explain its important features? [6M]
(b) Write the important differences and operation of continuous timer blocks and digital blocks? [6M]

(OR)

- (c) Explain about programmable system-on-chip processors and write its advantages? [6M]
(d) What are various modes of operation of ARM processor explain? [6M]

- 4(a) Define interrupt and Exceptions write their differences with an example? [6M]
(b) What is a serial port and parallel port device driver explain them? [6M]

(OR)

- (c) Explain about different types of interrupt handling schemes with example? [6M]
(d) Define dead lock condition and explain how it can be avoided? [6M]

- 5(a) Explain various types of bus architecture serial communication protocols? [6M]
(b) Explain about Ethernet importance and their protocols for wire less LAN's? [6M]

(OR)

- (c) What are different popular Network protocols used for wireless systems? [6M]
(d) Write a short notes on (i) SDMA (ii) IDMA [6M]

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M.Tech. I Semester Supplementary Examinations, October 2012

IMAGE AND VIDEO PROCESSING

(Signals and System Processing)

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal marks

- 1(a) Explain about sampling and quantization process of an image? [6M]
- (b) Explain indetail about Discrete cosine transform. [6M]
- (OR)**
- (c) Explain the elements of digital image processing system with a neat block diagram. [8M]
- (d) Discuss in detail about the advantages of colour images over monochrome images. [4M]

- 2(a) Explain in detail about smoothing spatial filter for image enhancement. [6M]
- (b) Explain about the edge based segmentation? [6M]
- (OR)**
- (c) Discuss indetail about region based segmentation. [6M]
- (d) Explain the operation Histogram equalization. [6M]

- 3(a) Explain about inter pixel redundancy and psycho visual redundancy. [6M]
- (b) Explain about runlength coding. [6M]
- (OR)**
- (c) Discuss the restoration of an image in the presence of noise. [6M]
- (d) Explain about wavelet coding. [6M]

- 4(a) Explain how video signal can be sampled. [6M]
- (b) Explain about three dimentional motion models? [6M]
- (OR)**
- (c) Explain various filtering operations on video signals. [6M]
- (d) Explain about photometric image formation. [6M]

- 5(a) Explain with neat diagram about region based motion estimation? [6M]
- (b) Explain how a wavelet transform is used for video coding. [6M]
- (OR)**
- (c) Explainnn about block matching algorithm and explain how it is used for 2-D motion estimation. [6M]
- (d) Discuss about multiresolution motion estimation. [6M]

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M.Tech. I Semester Supplementary Examinations, October 2012

MOBILE COMPUTING

(Software Engineering (IT))

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal marks

- 1(a) Explain about mobile computing architecture briefly. [6M]
(b) Explain about user mobility, bearer mobility and host mobility briefly. [6M]

(OR)

- (c) Explain about signal propagation in single path and multi path propagation [6M]
(d) Explain about frequency hopping spread spectrum [6M]
2(a) Explain about base station subsystem in GSM network. [6M]
(b) Explain the mechanism comprised by the radio interface. [6M]

(OR)

- (c) Explain about FDMA with neat diagram and what is the disadvantages of FDMA [6M]
(d) Explain about the persistence strategy and describe its types. [6M]
3(a) Explain about mobile IP registration and its types. [6M]
(b) Explain about IP-IN-IP encapsulation and draw the message format [6M]

(OR)

- (c) Explain about snooping TCP and explain its advantages and disadvantages. [6M]
(d) Explain about data transmission from MN to CN through FA with diagram [6M]
4(a) Describe about IEEE 802.11 protocol architecture briefly [6M]
(b) Explain about MAC layer of IEEE 802.11 briefly. [6M]

(OR)

- (c) Explain about ZRP routing protocol and describe its advantages and disadvantages. [6M]
(d) What is meant by route reply storms and packet salvaging. [6M]
5(a) Explain about class-0, class-1 and class-2 types of services provided by wireless transaction protocol (WTP) [6M]
(b) What are the services provided by wireless session protocol (WSP) [6M]

(OR)

- (c) Describe about the link management protocol [6M]
(d) Describe about LZCAP protocol [6M]

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M.Tech. I Semester Supplementary Examinations, October 2012

ADVANCED MATHEMATICS

(Thermal Engineering (MECH))

Time: 3 hours

Max. Marks: 60

Answer all the questions

All Questions carry equal marks

Note: For all numerical computation, three decimal accuracy is required
scientific and non programmable calculators are permitted.

- 1(a) Explain the gauss Elimination method of solving a system of linear equations. Apply this method and solve the following system

$$2x + y + z = 10$$

$$3x + 2y + 3z = 18$$

$$X + 4y + 9z = 16$$

[5+7M]

(OR)

- (b) Solve the following system of linear equation using LU – decomposition method

$$3x - 6y - 3z = -3$$

$$2x + 6z = -22$$

$$-4x + 7y + 4z = 3$$

[12M]

- 2(a) Find the largest Eigen value in modulus and the corresponding eigen vector of the following matrix using power method.

$$A = \begin{bmatrix} 15 & 4 & 3 \\ 10 & -12 & 6 \\ 20 & -4 & 2 \end{bmatrix}$$

[12M]

(OR)

- (b) Find the value of $\cos 1.74$ using the values given in the table below:

x :	1.70	1.74	1.78	1.82	1.86
Sin x :	0.9916	0.9857	0.9781	0.9691	0.9584

[12M]

- 3(a) Computer the values of $I = \int_0^1 \frac{dx}{1+x}$ by using simpson's 1/3 rule with 3,5 and 9 nodes. Improve your results by using Romberg integration. compare your result with the exact value. [6+5+1M]

(OR)

- (b) Solve the initial value problem

$$\frac{du}{dt} = -2tu^2, u(0) = 1$$

[12M]

With $h = 0.2$ on the interval $[0, 0.4]$ using the range-kutta (fourth order) method.

- 4(a) Solve the boundary value problem

$$u'' = u + x; u(0) = 0, u(1) = 0$$

with $h = \frac{1}{4}$ using the finite difference method of second order. [12M]

(OR)

- (b) Solve the boundary value problems

$$u'' = 6u^2; u(0) = 1, u(0.5) = 4/9$$

using shooting method: take the initial approximations:

$$u'(0) = \alpha_0 = -1.8 \text{ and } u'(0) = \alpha_1 = -1.9. \text{ Take } h = 0.1 \quad [12M]$$

- 5(a) State the Dirict problem and derive the five point difference equation for the square network ($h=k$). Deduce the corresponding equation for the poisson equation. [12M]

(OR)

- (b) Solve the equation $\nabla^2 \mu = -10(x^2 + y^2 + 10)$ over the square with sides $x = 0, y = 0, x = 3, y = 3$ with $\mu = 0$ on the boundary. Take the mesh length equal to one. [12M]

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M.Tech. I Semester Supplementary Examinations, October 2012

POWER CONVERTERS

(Power Electronics and Drives (EEE))

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal marks

- 1(a) Explain the 1- ϕ AC voltage controllers with PWM control with neat waveforms and also explain the effects of source and load inductances. [12M]

(OR)

- (b) Briefly explain 3- ϕ bidirectional delta-connected controllers with waveforms. [12M]

- 2(a) Write short notes in the following
(i) PWM control (ii) input power factor and Harmonic factor. [12M]

(OR)

- (b) (i) Briefly explain about 12 pulse converter
(ii) Evaluate the input power factor of a single-phase full converter. [12M]

- 3(a) Write short notes on the following
(i) Single phase single stage boost power factor corrected rectifier
(ii) Three phase boost PFC converter [12M]

(OR)

- (b) Briefly discuss the power circuit principle of operation [12M]

- 4(a) Explain about three phase inverter voltage control by
(i) Sinusoidal PWM (ii) 60° PWM [12M]

(OR)

- (b) With neat sketch explain about current source inverter. [12M]

- 5(a) Explain the principle of operation of diode-clamped multilevel inverters and write the features of it. [12M]

(OR)

- (b) Explain briefly about
(i) Concept of multilevel inverter
(ii) Features of flying-capacitor multilevel inverters [12M]

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M.Tech. I Semester Supplementary Examinations, October 2012

ADVANCED HEAT & MASS TRANSFER

(Thermal Engineering (MECH))

Time: 3 hours

Max. Marks: 60

Answer all the questions
All Questions carry equal marks

- 1(a) A 0.8m high and 1.5m wide double pane window consists of two 4mm thick layers of glass ($K = 78 \text{ W/mK}$) separated by a 10mm wide stagnant air space, $K = 0.026 \text{ W/mK}$. Determine the rate of heat transfer through this window and the temperature of the inside surface when the room is maintained at 20°C and the outside air is at -10°C . Take the convection heat transfer coefficient on the inside and outside surface of window as 10 and $40 \text{ W/W}^2\text{K}$ respectively? [6M]
- (b) Derive an expression for the temperature distribution in a hollow cylinder with heat generation? [6M]
- (OR)
- (c) A stainless steel Fin ($K = 20 \text{ W/mK}$) having diameter of 20 mm and length of 0.1m is attached to a wall at 300°C . The ambient temperature is 50°C and the heat transfer coefficient is $10 \text{ W/m}^2\text{K}$. The Fin tip is insulated. Determine (i) The rate of heat dissipation from the Fin (ii) The temperature at Fin tip (iii) The rate of heat transfer from the wall area covered by Fin if the Fin was not used? [6M]
- (d) A 25cm thick slab is made of material having thermal conductivity $40 \text{ W/m}^\circ\text{K}$. Measurements indicate that the temperature variation with in slab can be prescribed by the relation $t_x = 100 + 200x - 400x^2$ where 't' in $^\circ\text{C}$ and x is distance measure from one face in meters. Make calculation for temperature gradients and heat flow at planes $X = 0$, $X = 10\text{cm}$ and $X = 20 \text{ cm}$. [6M]
- 2(a) Define the terms the response of thermocouple and sensitivity of thermocouple and also derive an expression for time constant of the thermocouple? [6M]
- (b) The following data pertains to the junction of a thermocouple wire used to measure the temperature of gas stream. Density = 8500 kg/m^3 , specific heat is 325 J/kg K ; thermal conductivity is 40 W/mK and the heat transfer coefficient between the junction and gas is $215 \text{ W/m}^2 \text{ K}$. If thermocouple junction can be approximated as 1mm diameter sphere, determine how long it will take for the thermocouple to read 99% of the initial temperature difference? [6M]
- (OR)
- (c) A 2cm thick steel slab heated to 525°C is held in air stream having mean temperature of 25°C . Estimate the time interval when the slab temperature would not depart from the mean value of 25°C by more than 0.5°C at any point in the slab. The steel has the following thermo-physical properties $\rho = 7950 \text{ kg/m}^3$, $C_p = 455 \text{ J/kg K}$; $K = 46 \text{ W/mK}$; $h = 36 \text{ W/m}^2\text{K}$. [6M]
- (d) A short aluminum cylinder 5 cm diameter and 10 cm long is initially at uniform temperature of 200°C . It is suddenly subjected of a convection environment at 70°C and $h = 525 \text{ W/m}^2 \text{ }^\circ\text{C}$. Calculate the temperature at a radial position of 1.25 cm and a distance of 0.625 cm from one end of the cylinder 1 min after exposure to the environment? [6M]
- 3(a) The velocity distribution in the boundary layer of a flat plate is prescribed by the relation $\frac{\mu}{v_\infty} = \sin\left(\frac{\pi y}{2\delta}\right)$. Use momentum integral equation to develop an expression for the boundary layer thickness? [6M]

- (b) Neglecting viscous dissipation, the coefficient of convective heat transfer is anticipated depend upon the following parameters. Fluid viscosity ' μ ', fluid density ' ρ ', fluid thermal conductivity K , specific heat ' C_p ', flow velocity ' V ' and significant length ' l '. Consider the mass, length, time and temperature as the significant dimensions, set up a suitable correlation in terms of non-dimensional numbers for heat flow. Use Buckingham's π -theorem method of dimensional analysis? [6M]

(OR)

- (c) Water flows over a flat plate measuring $1\text{m} \times 1\text{m}$ with a velocity of 2ms^{-1} . The plate is at a uniform temperature of 90°C and the water temperature is 10°C . Estimate the length of plate over which the flow is laminar and the rate of heat transfer from the entire plate. The property of water at 50°C are $\rho = 988 \text{ kg/m}^3$; $\nu = 0.556 \times 10^{-6} \text{ m}^2/\text{sec}$; $Pr = 3.54$ and $K = 0.648 \text{ W/mK}$. [6M]
- (d) Lubricating oil at a temperature of 60°C enters a 1 cm diameter tube with a velocity 3.5 ms^{-1} . The tube surface is maintained at 30°C . Calculate the tube length required to cool the oil to 45°C . Assume that the oil has the following average properties for the temperature range of this problem. $\rho = 865 \text{ Kg/m}^3$, $K = 0.14 \text{ W/m}^\circ\text{K}$; $C_p = 1.78 \text{ kJ/kg}^\circ\text{K}$, $\nu = 9 \times 10^{-6} \text{ m}^2/\text{sec}$. [6M]
- 4(a) A small sphere (outside radius = 60mm) with a surface temperature of 300°C is located at the geometric centre of a large sphere (Inside diameter = 360 mm) with an inner surface temperature of 15°C . Calculate how much heat is emitted from the large sphere inner surface is incident upon the outer surface of the small sphere, assuming that both surfaces approach black body behavior. What is the net exchange of heat between two spheres? [6M]
- (b) A 100mm diameter disc ($\Sigma = 0.8$) at 800 K is at a distance of 2m from a disc ($\Sigma = 0.7$) of 2m diameter maintained at 300K . Find the net rate of radiant exchange? [6M]

(OR)

- (c) One hundred tubes of 1.27 cm diameter are arranged in a square array and exposed to atmospheric steam. Calculate the mass of steam condensed per unit length of tubes for a tube wall temperature of 98°C . [6M]
- (d) A 0.10 cm diameter and 15cm long wire has been laid horizontally and submerged in water at atmospheric pressure. The wire has a steady state voltage drop of 14.5V and a current of 4.25 A . Determine the heat flux and the excess temperature of the wire. The following equation applied for water boiling on a horizontal submerged surface $h = 1.54 (Q/A)^{0.75} = 5.58 (\Delta t)^3 \text{ W/m}^2\text{K}$. Where Q/A is the heat flux rate in W/m^2 and Δt is the temperature difference between surface and saturation? [6M]
- 5(a) Evaluate the mass transfer coefficient by dimensional analysis? [6M]
- (b) What is meant by mass transfer and what is the classification of mass transfer process? Explain. [6M]

(OR)

- (c) Air at 1 atm , 25°C containing small quantities of Iodine flow with a velocity of 5.18 ms^{-1} inside a 3.048 cm diameter tube. Determine the mass transfer coefficient for Iodine transfer from the gas stream to the wall surface. If C_m is the mean concentrate of Iodine in Kg mol/m^3 in the air stream, Determine the rate of deposition of Iodine on the tube surface where the Iodine concentration is zero? [6M]
- (d) The air pressure in a tyre tube of surface area 0.5m^2 and wall thickness of 0.01 m is approximated to drop from 2 bar to 1.99 bar in a period of 5 days . The solubility of air in rubber is 0.07 m^3 of air/ m^3 of rubber at 1 bar . Estimate the diffusivity of air in rubber at the operating temperature of 300 K . if the volume of air in tube is 0.025 m^3 ? [6M]